Highlights

Discrete Learning Algorithms for Precipitation Estimation from Commercial Microwave Links

Guy Even,Andreas Karrenbauer,Rex Lei,Jonatan Ostrometzky,Christian Sohler,Iván Soto

- Rainfall reconstruction from CMLs is formulated as an explicit inverse optimization problem.
- A benchmark is built from EURADCLIM rain maps and synthetic link attenuations.
- The formulation preserves path geometry instead of collapsing links to point measurements.

Discrete Learning Algorithms for Precipitation Estimation from Commercial Microwave Links

Guy Even[a], Andreas Karrenbauer[a], Rex Lei[a], Jonatan Ostrometzky[b], Christian Sohler[c] and Iván Soto[a]

[a] *MPI for Informatics, Saarbrücken, Germany*
[b] *School of Electrical Engineering, Tel-Aviv University, Tel Aviv, Israel*
[c] *University of Cologne, Cologne, Germany*

ARTICLE INFO

Keywords:
Commercial microwave links
Rainfall estimation
Inverse problems
Optimization
Remote sensing

ABSTRACT

Commercial microwave links (CMLs) are part of the infrastructure of wireless networks. Their measured attenuations have been studied as an opportunistic source for monitoring spatiotemporal rainfall and other atmospheric phenomena. CML attenuation measurements can enhance the spatiotemporal accuracy and resolution of existing weather monitoring instruments. In addition, they serve as stand-alone weather monitoring devices in places where dedicated weather monitoring devices are scarce or do not exist.

Current techniques for 2D rainfall map reconstruction usually reduce CML measurements to virtual rain gauges, namely point measurements, and rely on interpolation techniques such as inverse distance weighting or Kriging. While effective in many scenarios, these methods are suboptimal because they do not address the modeling error introduced when a path-integrated attenuation measurement is replaced by a single-point rain-intensity value.

In this study, we revisit the rainfall map reconstruction problem from CML signal attenuation measurements with a principled optimization approach. We formulate the problem of partial-to-complete field reconstruction as a physics-informed optimization problem. The reconstructed rainfall field is quantized and represented by pixel rainfall variables whose values are constrained to agree with the observed CML signal attenuations. The resulting solution minimizes a weighted sum of attenuation errors along the links, spatial differences between neighboring pixels, and the total rainfall over the map.

To evaluate our approach, we create a benchmark of hundreds of rainfall maps and CML configurations. Rainfall maps are extracted from EURADCLIM by identifying rain events over patches of approximately 50×50 km^2 across diverse terrain types and precipitation patterns. CMLs are overlaid using a publicly available Netherlands dataset, and attenuations are computed using the ITU-R P.838 model.

We then solve the inverse problem to reconstruct rainfall maps from attenuations and compare the results with inverse distance weighting. Overall, this study reframes rainfall reconstruction from opportunistic sensing networks as a well-posed inverse problem with an explicit objective function. The framework also provides interpretability benefits compared to purely data-driven approaches.

1. Introduction

GE:need complete rewrite

2. Introduction

Accurate rainfall estimation is a major tool used for accurate hydrology studies, weather prediction, flood forecasting, and urban drainage management. Reliable information on the spatial and temporal structure of precipitation is particularly important in urban and densely populated environments, where short-duration convective events may produce strong local gradients and rapid hydrological response. Conventional weather monitoring tools, primarily rain gauges and weather radars, provide valuable information but also have well-known limitations. Rain gauges offer direct point measurements, and are considered very accurate when properly installed, but their spatial density is often insufficient for resolving localized rainfall variability. Weather radars provide wide-area coverage, yet their

guyeven@mpi-inf.mpg.de (G. Even); andreas.karrenbauer@mpi-inf.mpg.de (A. Karrenbauer); rexlei@mpi-inf.mpg.de (R. Lei); jonatano@tauex.tau.ac.il (J. Ostrometzky); sohler@cs.uni-koeln.de (C. Sohler); ivanalisv@gmail.com (I. Soto)
ORCID(s):

estimates are not at ground level, and are prone to many uncertainties such as scattering, blockages, unaccounted vertical variability of rainfall, and limited near-ground representativeness.

Commercial microwave links (CMLs) offer a complementary source of rainfall information. These links are widely deployed as part of cellular telecommunication backhaul networks and continuously transmit electromagnetic signals between fixed antennas in a line-of-sight installments, nowadays at frequencies of 10 GHz to 90 GHz. During rainfall, raindrops affect the microwave radiation, causing measurable attenuation of the received signal. Since the frequencies range of operational links are sensitive to rain-induced attenuation ITU-R (2005); Gunn and East (1954); Olsen et al. (1978), the same infrastructure for communication was shown to also serve as an opportunistic environmental sensing network Overeem et al. (2013); Chwala and Kunstmann (2019); Fencl et al. (2015). As the CMLs are already deployed at large scale, often form dense networks in urban and suburban areas (especially in recent times, where dedicated CMLs networks for smart-city management are being deployed Ostrometzky and Messer (2024); Janco et al. (2022), and are gaining attention as part of the integrated sensing and communication (ISAC) paradigm Han et al. (2023)), they can provide near-ground measurements integrated along their paths, which span between tens of meters to several kilometers. Note that CMLs can be used in combination with satellite microwave links (SML) to further enhance their capabilities Nebuloni et al. (2025).

The physical basis for CML rainfall sensing is based on the attenuation of the signal that is caused by precipitation Gunn and East (1954); Olsen et al. (1978), and is described by the ITU-R power-law relation AK: We should cite the standard here, bu between rain rate and specific attenuation. Given the frequency and polarization of a link, the rain-induced attenuation accumulated along the link can be modeled as the path integral of the specific attenuation over the rainfall field. Thus, unlike a rain gauge, which measures rainfall at a point, a CML provides a path-integrated measurement. This distinction is fundamental. A single link does not directly observe the rainfall value at one location; rather, it constrains the average attenuation, and hence the rainfall intensity, along its propagation path. Consequently, rainfall reconstruction from CMLs is an inverse problem: the objective is to estimate a spatial rainfall field from a collection of line-integral-like measurements.

Many practical CML-based reconstruction methods simplify this inverse problem by first converting each link measurement into a single representative rain-rate value, often interpreted as a virtual rain gauge located at the link midpoint or at an assumed other point along the link path. Standard spatial interpolation techniques can then be applied to these virtual point observations. This approach is simple, computationally efficient, and often useful in practice. However, it also discards part of the geometric information contained in the link measurement. In particular, the fact that the attenuation is accumulated along an extended path is replaced by a point approximation. This simplification may introduce modeling errors, especially in heterogeneous rainfall fields or in regions where the geometry and density of the link network strongly influence the information available for reconstruction Gat and Messer (2019); Leijnse et al. (2008).

A more faithful treatment of CML data should account for the coupling between the unknown rainfall field and the measured attenuation along each link path. This coupling depends on the link geometry, length, frequency, polarization, and the spatial distribution of rain along the path. Reconstruction methods that explicitly incorporate this structure have the potential to use the available information more effectively, but they also lead to more challenging estimation problems. In particular, because the attenuation–rain-rate relation is nonlinear, the corresponding inverse problem may be formulated as a constrained nonconvex optimization problem over the rainfall field.

Evaluating such reconstruction methods is difficult in real operational settings because the true rainfall field is generally unknown. Real CML measurements also include additional effects that complicate the analysis, such as wet-antenna attenuation, baseline drift, quantization, hardware-dependent sampling, and measurement noise. While these effects are essential in operational systems, they make it difficult to isolate the intrinsic performance of the reconstruction algorithm itself. A controlled benchmark, in which the ground-truth rainfall field is known and the corresponding CML attenuations are generated through a forward physical model, provides a useful way to study this core inverse problem.

In this paper, we introduce such a benchmark framework. Starting from hourly EURADCLIM rainfall maps, we construct synthetic but physically grounded rain-event patches and generate corresponding noise-free CML attenuation measurements using the ITU-R forward model. This creates a controlled evaluation setting in which different reconstruction algorithms can be compared directly against the known rainfall field. By removing confounding measurement artifacts while preserving realistic spatial rainfall patterns, the proposed benchmark isolates the reconstruction problem and enables systematic analysis of the role of link geometry, interpolation assumptions, and optimization-based estimation.

Table 1
ITU-R P.838-3 coefficients for horizontal polarization and resulting specific attenuation γ_R for selected rain rates ($\mathrm{mm\,h^{-1}}$).

f (GHz)	κ_H	α_H	$\gamma_R(R=1)$ ($\mathrm{dB\,km^{-1}}$)	$\gamma_R(R=2)$ ($\mathrm{dB\,km^{-1}}$)	$\gamma_R(R=4)$ ($\mathrm{dB\,km^{-1}}$)	$\gamma_R(R=8)$ ($\mathrm{dB\,km^{-1}}$)
10	0.012	1.257	0.012	0.029	0.070	0.166
15	0.045	1.123	0.045	0.098	0.213	0.463
20	0.092	1.057	0.092	0.191	0.397	0.825
25	0.157	0.999	0.157	0.314	0.628	1.254
30	0.240	0.949	0.240	0.464	0.895	1.727
40	0.443	0.867	0.443	0.808	1.475	2.690

We use this framework to compare standard inverse-distance weighting (IDW), a link-aware extension tailored to CML measurements, and a nonconvex optimization formulation that directly couples the estimated rainfall field to the measured attenuations. AK: We should cite something for the other two methods In addition, we analyze reconstruction quality as a function of distance from the available links, thereby quantifying how the spatial arrangement of the CML network affects estimation accuracy.

2.1. Contributions

1. Introduction of a methodology for generating rain-event benchmarks for evaluation of rainfall estimation based on noise-free CML attenuation.
2. Creation of a benchmark (750 patches).
3. Introduction of a new algorithm ILDW - version of IDW tailored for CML's.
4. Introduction of rain-rate estimation as a nonconvex constrained optimization problem. Improve coupling between the estimation and measured attenuation.
5. Systematic evaluation of IDW, ILDW, and nonconvex optimization algorithms.
6. Evidence of quality of estimates as a function of distance from links.

3. Background and Related Work

3.1. ITU-R Forward Model: Rain-Induced Attenuation

During a rain event, the signal along a commercial microwave link experiences attenuation because raindrops absorb and scatter electromagnetic energy. We use Recommendation ITU-R P.838-3 for converting rain-rate into specific attenuation (ITU-R, 2005). The ITU-R model relates the rain rate R (in $\mathrm{mm\,h^{-1}}$) to the *specific attenuation* γ_R (in $\mathrm{dB\,km^{-1}}$) through the power law

$$\gamma_R = \kappa R^{\alpha}, \tag{1}$$

where the coefficients κ and α depend on the link frequency and polarization (ITU-R, 2005). See Table 1 for the coefficient values and resulting attenuation for horizontal polarization and rain rates of 1–8 $\mathrm{mm\,h^{-1}}$.

The total rain-induced attenuation is the integral of γ_R along the link. Thus, longer links accumulate more attenuation, and links operating at different frequencies or polarizations respond differently to the same rainfall field. For a horizontal terrestrial link of length L with uniform rain rate R along the path, the total rain-induced attenuation is therefore $A = \gamma_R \cdot L$. If the rain-rate field is discretized into pixels, the total rain-induced attenuation is the sum of the attenuations over the segments obtained by intersecting the link with the pixels it traverses.

4. Benchmark Construction GE:stable

In this section, we describe how we constructed our benchmark of rain-event patches combined with CML rain-attenuation. The resulting benchmark instances combine realistically derived rainfall maps with real-world link configurations, from which noise-free rain-induced attenuation measurements are generated for every link.

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 2
Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

4.1. Data Source

We extract rain events from the European climatological gauge-adjusted radar precipitation dataset EURADCLIM (Overeem et al., 2023). EURADCLIM is derived from OPERA European radar composites and provides hourly precipitation fields at 2 km spatial resolution over a large fraction of Europe. The dataset are distributed in HDF5 format according to the OPERA Data Information Model (ODIM) convention (EUMETNET OPERA, 2026). File-level access and documentation are provided through the KNMI data platform (KNMI, 2023).

4.2. Automatic Extraction of Rain Events

Each hourly EURADCLIM map is processed independently. We first average the rainfall field over a 4×4 window of native 2 km pixels, then retain the pixels whose smoothed rain rate is at least 3 mm h^{-1}. The retained pixels are partitioned into connected components under 8-neighborhood connectivity, and a component is accepted as a rain event when both side lengths of its axis-aligned bounding box fall between 50 km and 95 km. The accepted bounding box defines the extent of the benchmark patch, so patch dimensions follow the rain events themselves. Fixing a common patch size would instead require cropping the larger events or padding the smaller ones with dry area.

4.3. Refinement and Smoothing

We refine and smooth each rain patch derived from EURADCLIM rainfall maps by subdividing each pixel into 16×16 subpixels (each 125×125 m^2), followed by Gaussian smoothing with $\sigma = 1$ in subpixel units. The resulting rain patches are treated as the ground-truth rainfall field.

4.4. CML Dataset

CML metadata (including link locations, lengths, and frequencies) are obtained from the publicly available 4TU dataset (Overeem et al., 2016) (we assumed horizontal polarization for all links). For each rainfall patch, the set of links is cropped to the patch extent and overlaid on the rainfall field. The ITU-R rain-attenuation model is then applied along each link path to compute the corresponding attenuation values.

5. Benchmark Statistics

In this section, we summarize key properties of the benchmark patches and their associated link geometry. We describe the geographic distribution and size range of the selected rain events, illustrate representative patch instances, summarize link statistics and topology, and report how rainy and dry pixels are distributed with respect to distance from the third-closest link.

Patches.

Time. The rain patches were extracted from hourly EURADCLIM maps between 18 and 31 January 2023.

Geographic distribution. See Figure 1 for a depiction of the geographic locations of the 100 rain patches.

Patch sizes. A scatter diagram of patch side lengths is depicted in Figure 2. The smallest patch is 50×50 km and the largest is 88×88 km.

Patch statistics.

Sample patches. Sample rain-events from the benchmark are depicted in Figures 6 and 8 together with their geographic location.

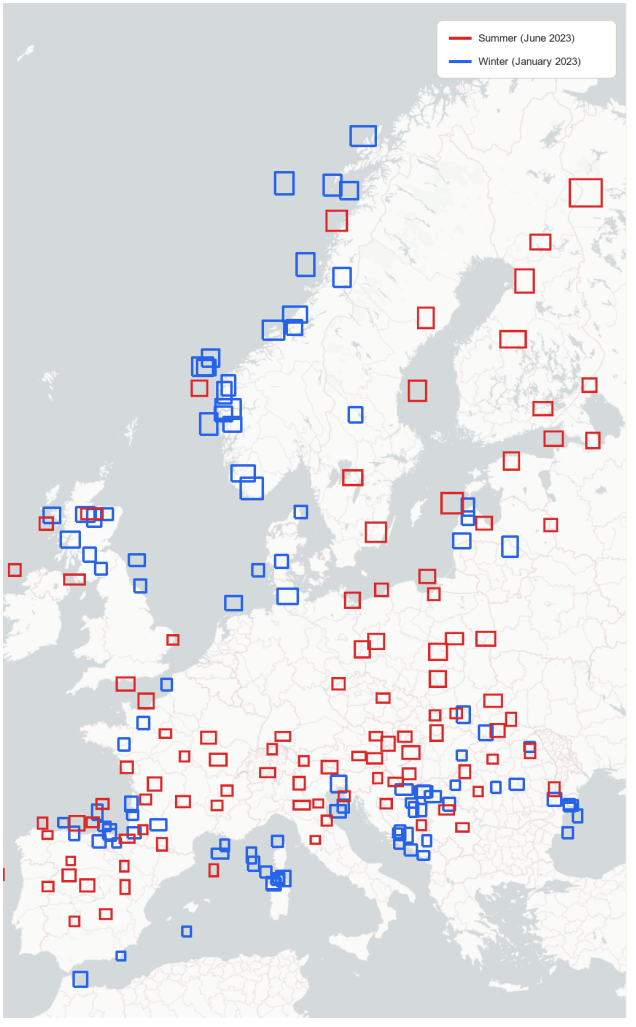


Figure 1: Geographic locations of the 100 benchmark patches.

Links. A depiction of the links in the largest 88×88 km patch appears in Figure 3. The pixels in this figure are binned according to their distance to the third-closest link. For a pixel p, let $d_3(p)$ denote the distance from the pixel center to the third-closest link. Equivalently, $d_3(p)$ is the smallest radius such that a disk centered at p intersects at least three links.

Table 4 provides data about the distribution of link lengths, frequencies and endpoint degrees in this patch (the degree of an endpoint is number of links incident to this endpoint). See Table 2 for the number of links and their multiplicities for typical patch sizes (multiplicity 2 means a link operating in two frequencies).

Rainy/Dry Pixel Distance Stratification. We considered all 100 benchmark patches and classified the pixels in each patch as rainy or dry using a rainfall threshold of $0.6\,\mathrm{mm\,h^{-1}}$. Each of these two pixel sets was then partitioned into bins according to the pixel-level distance $d_3(p)$. The mean and standard deviation of the rainy-pixel and dry-pixel counts in each distance bin, computed across all patches, are reported in Table 5.

6. Inverse Solvers: From Link Attenuations to Rainfall Maps

GE:stable

In this section, we present three solvers for the inverse problem: IDW, ILDW, and a nonconvex optimization model.

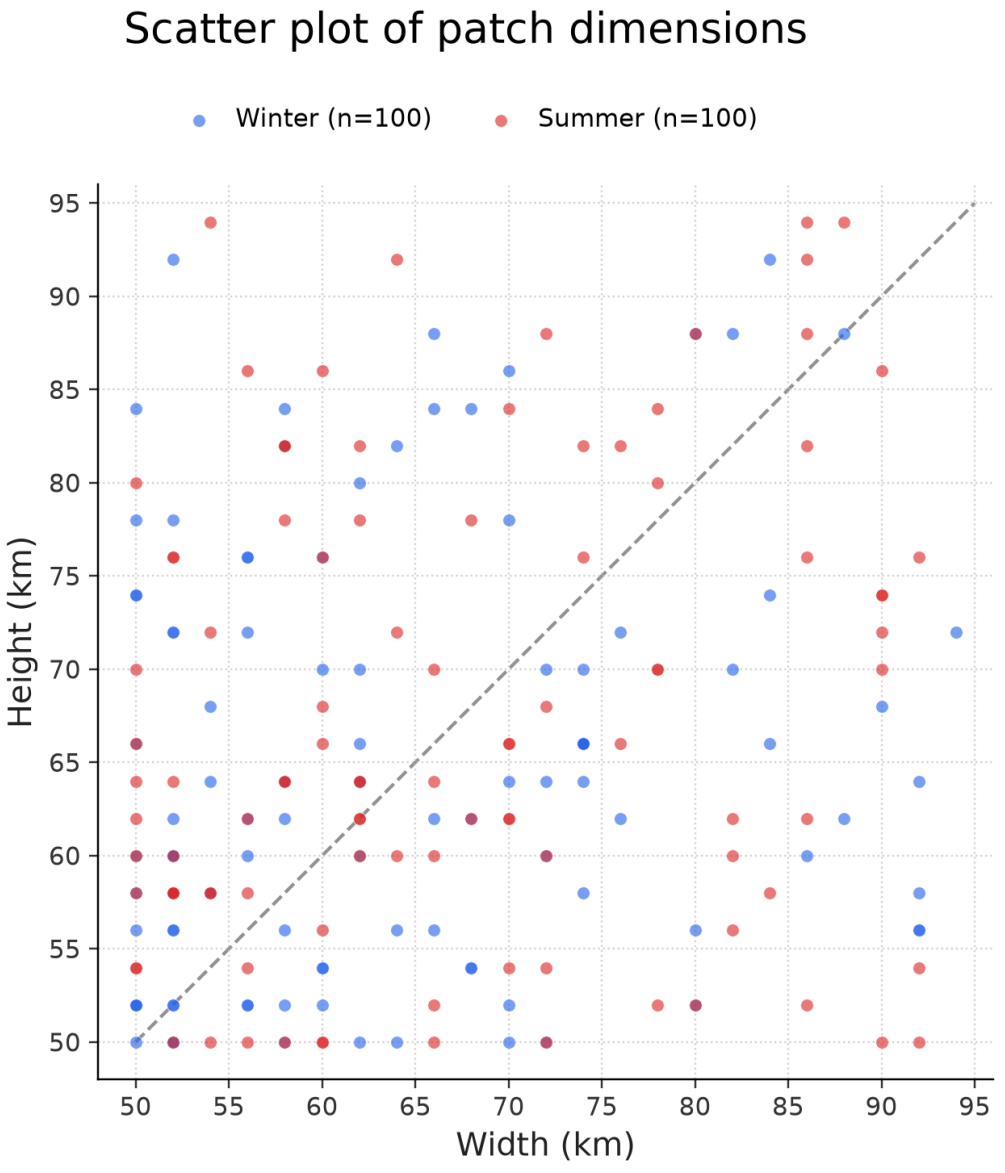


Figure 2: Patch width versus patch height for the 100 benchmark patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125 m).

6.1. Problem Formulation

The inputs for the inverse problem consists of a set of links $\mathcal{L}$ in a patch P (set of pixels). Every link ℓ has an attenuation A_ℓ and the following attributes: frequency, polarization, endpoints and length $|\ell|$.

The output is an estimated rain-rate field $\hat{R} : P \to \mathbb{R}_{\geq 0}$.

6.2. Rain Equivalent for Link Attenuation

The *rain-rate equivalent* R_ℓ of link ℓ is the uniform rain-rate along the link that would cause the link total attenuation A_ℓ. This rain-rate is derived from the ITU-R model by:

$$R_\ell \triangleq \left(\frac{A_\ell / |\ell|}{\kappa_\ell} \right)^{1/\alpha_\ell} ,$$

Descriptor	Mean	SD	Min	Median	Max
Wet fraction					
Mean wet rain rate ($\mathrm{mm\,h^{-1}}$)					
95th percentile rain rate ($\mathrm{mm\,h^{-1}}$)					
Coefficient of variation, wet pixels					
Area fraction above $5.000\,\mathrm{mm\,h^{-1}}$					
Relative gradient ($\mathrm{km^{-1}}$)					

Table 3
Rain-pattern descriptors across the 100 benchmark patches. Each patch contributes one value per descriptor, computed on its ground-truth rainfall field. Wet pixels are those at or above $0.6\,\mathrm{mm\,h^{-1}}$. The coefficient of variation is the standard deviation of the wet-pixel rain rates divided by their mean, and the relative gradient is the mean gradient magnitude of the field divided by its mean rain rate.

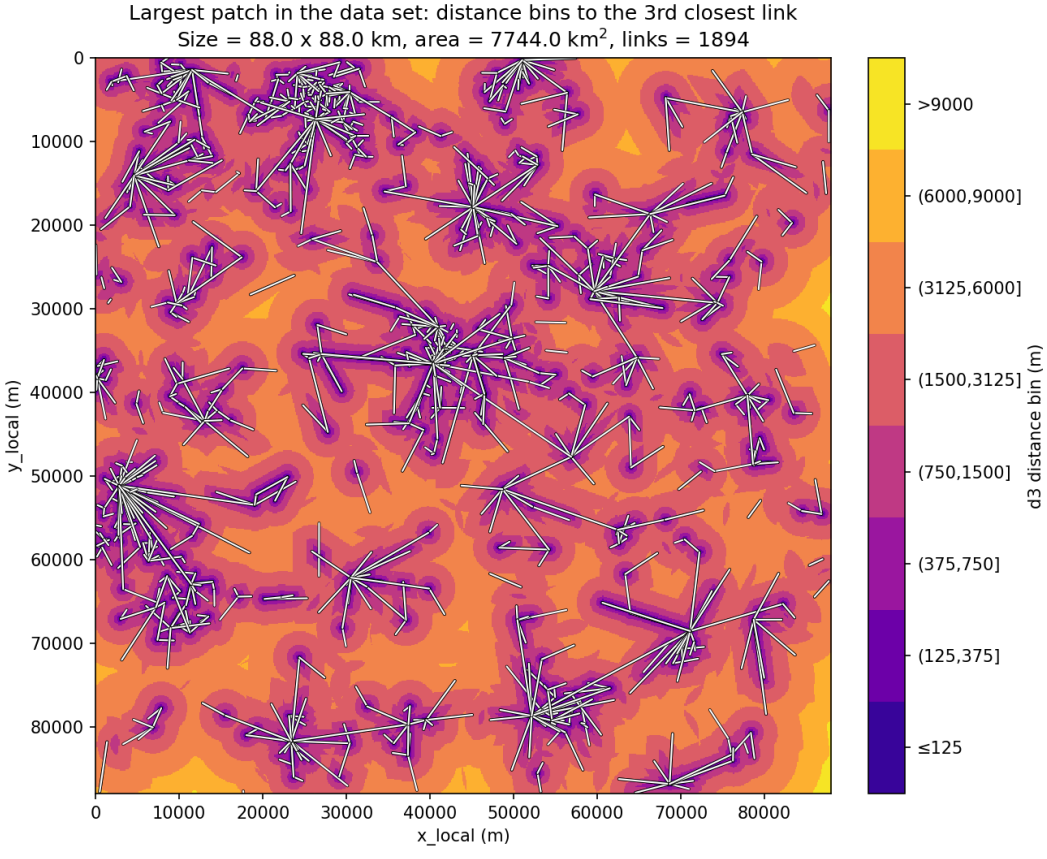


Figure 3: d_3-bin map for the largest evaluated patch in the benchmark. Each pixel is colored according to its d_3 bin, and the complete link geometry is overlaid in white.

where $(\kappa_\ell, \alpha_\ell)$ the ITU-R P.838-3 coefficients corresponding to the link's frequency and polarization.

6.3. IDW

Inverse-distance weighting (IDW) is one of the standard interpolation strategies used in commercial-microwave-link rainfall reconstruction (Zimmerman et al., 1999; Guo-Dong et al., 2003; Chen and Liu, 2012; Chwala and Kunstmann, 2019; Eshel et al., 2020, 2021). The input consists of a set of links, where each link ℓ carries a total attenuation, frequency, and polarization.

Length bin (km)	Count	%
[0, 1)	330	17.4
[1, 2)	515	27.2
[2, 4)	546	28.8
[4, 8)	387	20.4
[8, 16)	109	5.8
[16, 24]	7	0.4

Frequency bin (GHz)	Count	%
[10, 15)	20	1.1
[15, 20)	67	3.5
[20, 25)	79	4.2
[25, 30)	257	13.6
[30, 35)	84	4.4
[35, 40)	1306	69.0
[40, 60]	81	4.3

Endpoint-degree bin	Count	%
1	168	15.0
2	560	49.9
3, 4	246	21.9
5, 6	68	6.1
7..10	39	3.5
11..20	26	2.3
≥ 21	16	1.4

Table 4
Link statistics for the largest-patch. The first panel counts links by path-length interval, the second counts links by carrier-frequency interval, and the third counts degrees of link endpoints.

	Rainy pixels			Non-rainy pixels		
d_3 bin (m)	Mean count	Mean %	SD	Mean count	Mean %	SD
[0, 125]	4267	1.790	960	564	1.840	778
(125, 375]	16 738	7.010	3803	2261	7.400	3094
(375, 750]	26 810	11.220	6707	3389	11.090	4710
(750, 1500]	51 989	21.760	14 468	6447	21.090	9258
(1500, 3125]	88 839	37.180	27 531	10 776	35.250	15 554
(3125, 6000]	45 929	19.220	15 679	6197	20.270	8990
(6000, 9000]	3836	1.610	2726	850	2.780	1290
(9000, ∞)	514	0.220	967	86	0.280	251
Total	238 922	100.000		30 569	100.000	

Table 5
Ground-truth rainy and non-rainy pixel counts per d_3 bin. Values are mean and standard deviation across patches.

IDW for CML's replaces every link ℓ with a virtual rain-gauge g_ℓ positioned at the midpoint between the endpoints of ℓ. The rain-rate associated with g_ℓ is simply the rain-rate equivalent R_ℓ. Inverse-distance weighting is applied with the virtual gauges positions and rain-rates as inputs.

We consider a setting in which the weight of a gauge decreases quadratically (i.e., decay exponent equals 2) and the maximum radius of influence $r_{\max}$ is set to 15 km. Distances to pixels are measured to the center of the pixel. If one or more link midpoints lie inside pixel p, then the estimated rainfall in pixel p is set to the mean of their rainfall estimates. The pseudocode for IDW appears in Algorithm 1.

Algorithm 1: Discretized IDW. Each link is replaced with a virtual gauge located at the midpoint of the link. The rain-rate at the gauge is set to the rain-rate equivalent of its link. IDW is then applied with respect to these virtual gauges. We set the exponent of the decay to 2.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R} : P \to \mathbb{R}_{\geq 0}$

foreach *link* $\ell \in \mathcal{L}$ **do** `// compute rain-rate equivalent per link`
  $(\ell.\kappa, \ell.\alpha) \leftarrow \mathrm{ITU838}(\ell.\mathrm{freq}, \ell.\mathrm{polarization})$
  $R_\ell \leftarrow \left(\frac{A_\ell / |\ell|}{\ell.\kappa} \right)^{1/\ell.\alpha}$
foreach *pixel* $p \in P$ **do** `// compute rain` $\hat{R}(p)$
  $M_p \leftarrow \{\ell \in \mathcal{L} : \ell.\mathrm{midpoint} \in p\}$
  if $M_p \neq \emptyset$ **then**
    $\hat{R}(p) \leftarrow \frac{1}{|M_p|} \sum_{\ell \in M_p} R_\ell$ `// pixel contains a link endpoint`
  else
    $N_p \leftarrow \{\ell \in \mathcal{L} : \|\ell.\mathrm{midpoint} - p.\mathrm{center}\|_2 \leq r_{\max}\}$
    if $N_p = \emptyset$ **then**
      $\hat{R}(p) \leftarrow 0$ `// pixel far from links`
    else
      $\hat{R}(p) \leftarrow \sum_{\ell \in N_p} \frac{\|\ell.\mathrm{midpoint} - p.\mathrm{center}\|_2^{-2}}{\sum_{\ell' \in N_p} \|\ell'.\mathrm{midpoint} - p.\mathrm{center}\|_2^{-2}} \cdot R_\ell$

6.3.1. ILDW

We propose a variation of IDW that alleviates the inherent mismatch between the measured and IDW-estimated link attenuation (see **??**). We refer to this variation as inverse link-distance weighting (ILDW). ILDW assigns a weight to every link that is inversely proportional to the distance between the pixel and the link, where

$$\mathrm{dist}(p, \ell) \triangleq \min_{q \in \ell} \|p.\mathrm{center} - q\| \ .$$

As in IDW, we set the decay exponent to 2 and the maximum radius of influence $r_{\max}$ is set to 15 km. If one or more links cross the pixel, the estimated rain-rate is taken to be the mean of the corresponding link rain-rate equivalents. Otherwise, the estimate is a weighted average over links within distance $r_{\max}$ (set to 15km). The pseudocode for ILDW is shown in Algorithm 2, with lines that differ from IDW highlighted in red.

Algorithm 2: Discretized ILDW. Rainfall is averaged by inverse squared distance to a link. Lines that are different from their corresponding lines in IDW are colored red.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R} : P \to \mathbb{R}_{\geq 0}$

foreach *link* $\ell \in \mathcal{L}$ **do** // compute rain-rate equivalent per link
  $(\ell.\kappa, \ell.\alpha) \leftarrow \mathrm{ITU838}(\ell.\mathrm{freq}, \ell.\mathrm{polarization})$
  $R_\ell \leftarrow \left(\frac{A_\ell / |\ell|}{\ell.\kappa} \right)^{1/\ell.\alpha}$
foreach *pixel* $p \in P$ **do** // compute rain $\hat{R}(p)$
  $X_p \leftarrow \{\ell \in \mathcal{L} : \ell \cap p \neq \emptyset\}$
  if $X_p \neq \emptyset$ **then**
    $\hat{R}(p) \leftarrow \frac{1}{|X_p|} \sum_{\ell \in X_p} R_\ell$ // at least one link traverses the pixel
  else
    $N_p \leftarrow \{\ell \in \mathcal{L} : \mathrm{dist}(p, \ell) \leq r_{\max}\}$
    if $N_p = \emptyset$ **then**
      $\hat{R}(p) \leftarrow 0$ // pixel far from links
    else
      $\hat{R}(p) \leftarrow \sum_{\ell \in N_p} \frac{\mathrm{dist}^{-2}(p, \ell)}{\sum_{\ell' \in N_p} \mathrm{dist}^{-2}(p, \ell')} \cdot R_\ell$

6.3.2. Nonconvex Formulation

We formulate the inverse problem as a constrained optimization problem. The decision variables are the estimated rain-rates $\hat{R}(p)$ for all pixels $p \in P$. The constraints are simply nonnegativity constraints that reflect the physical requirement that rainfall is nonnegative. The objective is to minimize $J(\hat{R})$, where the cost function J consists of the 4 weighted terms:

$$J(\hat{R}) = w_{\text{atten}} \cdot J_{\text{atten}}(\hat{R}) + w_{1d} \cdot J_{1d}(\hat{R}) + w_{2d} \cdot J_{2d}(\hat{R}) + w_{\text{total}} \cdot J_{\text{total}}(\hat{R}) \,.$$

The terms in $J(\hat{R})$ are as follows:

1. The attenuation term J_{atten} quantifies disagreement between observed and estimated link attenuations. For a rain-rate field estimate $\hat{R} : P \to \mathbb{R}_{\geq 0}$ and a link ℓ, let $A_\ell(\hat{R})$ denote the attenuation induced on ℓ by $\hat{R}$. This attenuation is computed using the ITU-R forward model with the link's frequency and polarization. The attenuation term J_{atten} is defined by

$$J_{\text{atten}}(\hat{R}) \triangleq \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{\left(A_\ell(\hat{R}) - A_\ell\right)^2}{|\ell|} \,.$$

2. The first-difference term J_{1d} enforces spatial smoothness, and is defined by

$$J_{1d}(\hat{R}) \triangleq \frac{1}{|P|} \sum_{(u,v) \in \mathcal{E}(P)} (\hat{R}(u) - \hat{R}(v))^2 \,,$$

   where $\mathcal{E}(P)$ is the set of horizontally and vertically adjacent pixel pairs in patch P.
3. The second-difference term J_{2d} penalizes curvature, and is defined by

$$J_{2d}(\hat{R}) = \frac{1}{|P|} \sum_{(p,q,t) \in \mathcal{T}(P)} \left((\hat{R}(q) - \hat{R}(p)) - (\hat{R}(t) - \hat{R}(q))\right)^2,$$

   where $\mathcal{T}(P)$ is the set of horizontal and vertical triples of collinear pixels in patch P.
4. The shrinkage term J_{total} biases the solution toward smaller overall rainfall values, and is defined by

$$J_{\text{total}}(\hat{R}) \triangleq \frac{1}{|P|} \sum_{p \in P} \hat{R}(p)^2.$$

Implementation Details:

1. The weight of each term is set to the reciprocal of its value evaluated at the ILDW solution. We further divide w_{2d} by 2 and w_{total} by $|P|$. This normalizes the objective instance-by-instance.
2. We use the L-BFGS-B algorithm (Byrd et al., 1995; Zhu et al., 1997), a limited-memory quasi-Newton method for smooth constrained optimization, as implemented in SciPy (Virtanen et al., 2020).

Nonconvex Cost Function: The ITU-R forward model (Equation (1)) attaches an attenuation exponent α to every link based on its frequency and polarization. When $\alpha = 1$, the predicted attenuation is linear in the rain-rate variables, so J_{atten} is a sum of squares of linear residuals and is convex. For $\alpha \neq 1$, the power-law attenuation makes the residual nonlinear, and the attenuation term is generally nonconvex. The nonconvexity emphasizes the importance of the initial solution used by the solver because local minima need not be global minima. We illustrate this nonconvexity with a concrete 3×3 single-link example in Appendix A.1.

Summary of the Nonconvex Inverse Solver. The input to the inverse solver is a patch P and a set $\mathcal{L}$ of links. We treat each link $\ell \in \mathcal{L}$ as a record containing its geometry and physical attributes that include its frequency f_ℓ, polarization, length $|\ell|$, and measured attenuation A_ℓ. The L-BFGS-B solver also takes an initial rain-rate field $\hat{R}_{\text{init}}$ as input. We write

$$\hat{R} \leftarrow \text{LBFGSB}(\mathcal{L}, P, \hat{R}_{\text{init}})$$

to denote one invocation of the solver and the resulting rain-rate field estimate.

Coping with nonconvexity. We use three strategies to deal with the nonconvexity of J_{atten}:

1. Use ILDW as the initial solution. We refer to this solver as SOLVER(ILDW).
2. We initialize the nonconvex solver with the ground-truth rainfall field. This configuration, denoted SOLVER(GT), serves as a best-case reference for the nonconvex formulation.
3. Obtain a convex surrogate objective by replacing each link with a virtual link that keeps the same geometry but changes its frequency to f^0, chosen so that the attenuation exponent equals 1 (i.e., $f^0 = 25\,\text{GHz}$ for horizontal polarization). We refer to this solver as CONVEX SOLVER.
4. Apply homotopy continuation to smoothly transition from the convex surrogate to the original nonconvex formulation. We refer to this solver as HOMOTOPY SOLVER.

We elaborate on the last two strategies below.

6.4. CONVEX SOLVER: Convex Surrogate Inverse Solver

We construct a virtual link set $\mathcal{L}^0$ from the original link set $\mathcal{L}$. For each link $\ell \in \mathcal{L}$, the corresponding virtual link $\ell^0 \in \mathcal{L}^0$ keeps the same geometry and length, but uses the frequency f^0 for which the ITU-R attenuation exponent equals 1.[1] The attenuation of the virtual link ℓ^0 is set to

$$A_\ell^0 \triangleq \kappa_0 \cdot R_\ell \cdot |\ell|,$$

where κ_0 is the ITU-R coefficient at frequency f^0 and R_ℓ is the rain-rate equivalent of the original link ℓ (see Section 6.2). For $\mathcal{L}^0$, the attenuation error term J_{atten} is a sum of squares and is therefore convex. We write

$$\hat{R}^0 \leftarrow \text{LBFGSB}(\mathcal{L}^0, P, \hat{R}_{\text{init}})$$

to denote applying L-BFGS-B to the convex surrogate inverse problem. The surrogate objective is convex and includes the strictly convex J_{total}, so it has a single local minimum, which is also the global minimum. Thus, the initial solution $\hat{R}_{\text{init}}$ affects only the convergence speed of L-BFGS-B (In our implementation, $\hat{R}_{\text{init}} = 0.6$ for all pixels.). We refer to the resulting solver as CONVEX SOLVER.

6.5. HOMOTOPY SOLVER: Smooth Transition from Convex to Nonconvex

We extend the virtual-link construction to a parameter $\lambda \in [0, 1]$. For each λ, let

$$\mathcal{L}^\lambda \triangleq \{\ell^\lambda : \ell \in \mathcal{L}\}.$$

The virtual link ℓ^λ keeps the geometry, length, and polarization of ℓ, but has frequency

$$f_\ell^\lambda \triangleq (1-\lambda) f^0 + \lambda f_\ell.$$

Let $(\kappa_\ell(\lambda), \alpha_\ell(\lambda))$ be the ITU-R coefficients at frequency f_ℓ^λ. The attenuation of ℓ^λ is set to

$$A_\ell^\lambda \triangleq |\ell| \cdot \kappa_\ell(\lambda) \cdot R_\ell^{\alpha_\ell(\lambda)}.$$

Algorithm 3: HOMOTOPY SOLVER- a solver that applies a smooth transition from convex optimization to optimizing the nonconvex objective. Assume the same polarization for all the links.

Data: link set $\mathcal{L}$, patch P, initial field $\hat{R}_{\text{init}}$, number of homotopy steps n
Result: rainfall field $\hat{R} : P \to \mathbb{R}_{\geq 0}$

$\hat{R}^0 \leftarrow \text{LBFGSB}(\mathcal{L}^0, P, \hat{R}_{\text{init}})$
for $i = 1, \ldots, n$ **do**
  $\hat{R}^{i/n} \leftarrow \text{LBFGSB}(\mathcal{L}^{i/n}, P, \hat{R}^{(i-1)/n})$
return $\hat{R}^1$

[1] Note that f^0 depends on the link's polarization. For simplicity, we consider a single polarization.

7. Experimental Results and Analysis

We executed the three inverse solvers: IDW, ILDW, and the three versions of nonconvex solver on 100 rain-events patches in the benchmark. The ground-truth in these experiments is the rain-rate fields of the benchmark (from which link attenuations were derived).

7.1. Notation

For a patch P and solver s, let $R_P(p)$ denote the ground-truth for pixel p. Let $\widehat{R}_{P,s}(p)$ denote the estimate returned by solver s for the pixel $p \in P$. Define the means:

$$\overline{\widehat{R}}_{P,s} \triangleq \frac{1}{|P|}\sum_{p\in P}\widehat{R}_{P,s}(p), \qquad \overline{R}_P \triangleq \frac{1}{|P|}\sum_{p\in P}R_P(p).$$

For a patch P and solver s, we compute three values: the root mean squared error

$$\mathrm{RMSE}_{P,s} \triangleq \sqrt{\frac{1}{|P|}\sum_{p\in P}\left(\widehat{R}_{P,s}(p) - R_P(p)\right)^2},$$

the signed bias

$$\mathrm{Bias}_{P,s} \triangleq \frac{1}{|P|}\sum_{p\in P}\left(\widehat{R}_{P,s}(p) - R_P(p)\right),$$

and the Pearson correlation coefficient

$$r_{P,s} \triangleq \frac{\sum_{p\in P}\left(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s}\right)\left(R_P(p) - \overline{R}_P\right)}{\sqrt{\sum_{p\in P}\left(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s}\right)^2}\sqrt{\sum_{p\in P}\left(R_P(p) - \overline{R}_P\right)^2}}.$$

For a pixel p, the absolute error (AE) is defined by

$$AE_{P,s}(p) \triangleq |\widehat{R}_{P,s}(p) - R_P(p)|,$$

and the relative absolute error (RAE) is defined by

$$RAE_{P,s}(p) \triangleq \frac{AE_{P,s}(p)}{R_P(p)}.$$

To avoid division by zero, we use the absolute error for nonrainy pixels.

7.2. RMSE, Bias, and Pearson Correlation Results and Analysis

The mean and standard deviation of these statistics stratified by d_3 bins are summarized in **??**. The SOLVER(GT) results provide a best-case reference for the nonconvex formulation when the solver is initialized with the ground truth. The main pattern is that errors increase with distance from the CML network for all methods, but the increase is much weaker for CONVEX SOLVER than for IDW. Across all d_3 bins, CONVEX SOLVER has lower mean RMSE than IDW, with relative reductions of about 28%–44%. The improvement is largest in the near-link and intermediate-distance bins, where CONVEX SOLVER also raises Pearson r from 0.967 to 0.989 in the closest bin and from 0.844 to 0.932 in the (1500, 3125] m bin. At larger distances, IDW develops a positive bias, reaching 0.63–0.71 $\mathrm{mm\,h^{-1}}$ beyond 6000 m, whereas CONVEX SOLVER reduces this bias to 0.33 and 0.24 $\mathrm{mm\,h^{-1}}$.[2] Thus, the convex formulation improves both error magnitude and bias relative to IDW, especially away from the immediate vicinity of the links. After rounding, the statistics of CONVEX SOLVER and HOMOTOPY SOLVER are the same.

[2]The last distance bins contain relatively few pixels, so their statistics should be interpreted more cautiously.

Method	n	RMSE	Bias	Pearson r
IDW	200	2.18 ± 1.56	0.05 ± 0.32	0.835 ± 0.071
ILDW	200	2.07 ± 1.50	0.07 ± 0.28	0.849 ± 0.065
SOLVER(ILDW)	200	1.94 ± 1.54	0.06 ± 0.27	0.875 ± 0.074
CONVEX SOLVER	200	1.53 ± 1.17	0.05 ± 0.13	0.918 ± 0.045
HOMOTOPY SOLVER	200	1.53 ± 1.17	0.04 ± 0.13	0.918 ± 0.045
SOLVER(GT)	200	0.61 ± 0.49	0.01 ± 0.03	0.973 ± 0.031

Table 6
Patchwise reconstruction metrics across the full 200-patch benchmark (100 winter and 100 summer patches). Each entry is the mean ± sample SD of one value per patch.

Comparison	n	wins A:IDW	mean A	mean IDW	mean diff	SD diff	SE	t	one-sided p	99% upper bound	decision
ILDW- IDW	100	97:3	1.24	1.31	-0.07	0.09	0.01	-7.86	2.42×10^{-12}	-0.05	reject
SOLVER(ILDW)- IDW	100	98:2	1.06	1.31	-0.25	0.21	0.02	-11.83	3.89×10^{-16}	-0.20	reject
CONVEX SOLVER- IDW	100	100:0	0.85	1.31	-0.46	0.50	0.05	-9.30	1.17×10^{-15}	-0.34	reject
HOMOTOPY SOLVER- IDW	100	100:0	0.85	1.31	-0.46	0.50	0.05	-9.29	1.83×10^{-15}	-0.34	reject
SOLVER(GT)- IDW	100	100:0	0.50	1.31	-0.80	1.09	0.11	-7.41	2.20×10^{-11}	-0.55	reject

Table 7
Paired t comparisons for patchwise RMSE differences relative to the IDW baseline. Negative differences indicate lower RMSE for method A. The wins column reports the number of patches for which A or IDW has lower RMSE. No ties occurred. Tests are one-sided and evaluated at $\alpha = 0.01$.

Comparison	n	wins A:IDW	mean $\lvert\text{Bias}_A\rvert$	mean $\lvert\text{Bias}_{\text{IDW}}\rvert$	mean diff	SD diff	SE	t	one-sided p	99% upper bound	decision
ILDW- IDW	100	67 : 33	0.14	0.16	−0.02	0.04	0.004	−4.77	3.14×10^{-6}	−0.01	reject
SOLVER(ILDW)- IDW	100	77 : 23	0.10	0.16	−0.05	0.11	0.011	−5.20	5.37×10^{-7}	−0.03	reject
CONVEX SOLVER- IDW	100	78 : 22	0.05	0.16	−0.11	0.21	0.021	−5.23	4.67×10^{-7}	−0.06	reject
HOMOTOPY SOLVER- IDW	100	78 : 22	0.05	0.16	−0.11	0.21	0.021	−5.24	4.62×10^{-7}	−0.06	reject
SOLVER(GT)- IDW	100	85 : 15	0.02	0.16	−0.14	0.26	0.026	−5.38	2.52×10^{-7}	−0.08	reject

Table 8
One-sided paired mean comparisons for patchwise absolute-bias differences relative to the IDW baseline. Negative differences indicate that method A is closer to zero bias than IDW. The wins column reports the number of patches for which A or IDW has smaller absolute bias. No ties occurred. Tests are one-sided and evaluated at $\alpha = 0.01$.

Paired t analysis for RMSE

Table 7 summarizes the paired t tests for patchwise RMSE differences relative to IDW. Negative differences indicate that method A has lower RMSE than IDW, which is the one-sided alternative tested here. The results show that all alternatives improve significantly over IDW. The improvement from ILDW is modest but highly consistent, with lower RMSE in 97 of the 100 patches and a mean reduction of 0.07 mm h^{-1}. The optimization-based methods show larger gains. In particular, CONVEX SOLVER and HOMOTOPY SOLVER improve over IDW in all 100 patches, with a mean RMSE reduction of 0.46 mm h^{-1}.

Most importantly, the 99% one-sided upper confidence bound for CONVEX SOLVER is -0.34 mm h^{-1}. Thus, even under this conservative bound, the mean patchwise RMSE reduction relative to IDW is at least 0.34 mm h^{-1}.

Paired t analysis for absolute bias

Table 8 summarizes the paired mean tests for patchwise absolute-bias differences relative to IDW. Negative differences indicate that method A is closer to zero bias than IDW, which is the one-sided alternative tested here. The results show that all alternatives significantly reduce absolute bias relative to IDW. The reduction is more heterogeneous than for RMSE, as indicated by the win counts. In particular, CONVEX SOLVER and HOMOTOPY SOLVER have smaller absolute bias than IDW in 78 of the 100 patches, and reduce the mean absolute bias from 0.16 to 0.05 mm h^{-1}.

Paired comparison of Pearson correlation

Table 9 summarizes patchwise wins for Pearson correlation relative to IDW, where higher r indicates better spatial agreement with the ground-truth rainfall field. The results show that all alternatives improve this measure in most patches. The evidence is strongest for CONVEX SOLVER, HOMOTOPY SOLVER, and SOLVER(GT), which have higher r than IDW in all 100 patches. This indicates that the optimization-based methods improve not only RMSE and bias, but also the spatial agreement of the reconstructed rainfall fields.

7.3. Confusion Summaries

Table 10 summarizes the wet/dry classification performance of the methods at the 0.6 mm h^{-1} threshold. The SOLVER(GT) results provide a best-case reference for the nonconvex formulation when the solver is initialized with the ground truth. Relative to IDW, CONVEX SOLVER reduces false-wet classifications from 4.0% to 2.3% of all pixels, at the cost of increasing false-dry classifications from 0.2% to 0.4%. The conditional rates show a similar pattern: the

Comparison	wins A:IDW
ILDW- IDW	94 : 6
SOLVER(ILDW)- IDW	98 : 2
CONVEX SOLVER- IDW	100 : 0
HOMOTOPY SOLVER- IDW	100 : 0
SOLVER(GT)- IDW	100 : 0

Table 9
Patchwise Pearson correlation wins relative to the IDW baseline. The wins column reports the number of patches in which method A or IDW has higher r. No ties occurred.

Method	n	Fraction of all pixels				Conditional rate			
		TP	FP	TN	FN	TP	FP	TN	FN
IDW	200	0.799	0.082	0.116	0.004	0.994	0.488	0.442	0.006
ILDW	200	0.798	0.076	0.122	0.004	0.994	0.456	0.474	0.006
SOLVER(ILDW)	200	0.798	0.072	0.125	0.004	0.994	0.404	0.526	0.006
CONVEX SOLVER	200	0.797	0.047	0.150	0.005	0.992	0.307	0.623	0.008
HOMOTOPY SOLVER	200	0.797	0.047	0.150	0.005	0.992	0.308	0.622	0.008
SOLVER(GT)	200	0.800	0.013	0.185	0.003	0.996	0.176	0.754	0.004

Table 10
Patch-averaged wet/dry confusion at the 0.6 mm/h threshold across all 200 patches. Fraction columns are normalized by each patch's pixel count; conditional TP/FN rates use wet GT pixels and FP/TN rates use dry GT pixels.

false-wet rate over dry ground-truth pixels decreases from about 0.6 for IDW to 0.4 for CONVEX SOLVER, while the false-dry rate over wet ground-truth pixels remains below 0.01 for all methods.

7.4. Attenuation Error

Table 11 summarizes the error between the ground-truth attenuation and the attenuation reconstructed from the estimated rainfall field using the forward model. The main quantity is attenuation MAE, reported in dB km^{-1}. The table also reports the attenuation cost-function term J_{atten}. For a given patch $\boldsymbol{P}$ and solver s, the attenuation MAE per kilometer equals $\frac{1}{|\mathcal{L}_P|}\sum_{\ell\in\mathcal{L}_P}\frac{\left|\hat{A}_{\ell,s}-A_\ell\right|}{|\ell|}$, where $\hat{A}_{\ell,s}$ is the attenuation implied on link ℓ by the ITU-R forward model according to the rain-rate solution $\hat{\boldsymbol{R}}_{P,s}$, A_ℓ is the ground-truth attenuation, and $|\ell|$ is the link length in kilometers.

The results show that the optimization-based methods fit the link attenuations much more closely than the interpolation baselines. CONVEX SOLVER reduces the attenuation MAE from 0.050 for IDW and 0.014 for ILDW to 1.1×10^{-3} dB km^{-1}, while HOMOTOPY SOLVER gives the smallest mean value, 7.4×10^{-4} dB km^{-1}. The same ordering is reflected in J_{atten}, as expected because this term is part of the cost function. GE: discuss the error compared to ITU attenuation per km for various

7.5. Median RAE for wet pixels and median AE for dry pixels

Because pixel-level errors within a patch are spatially dependent, we do not treat individual pixels as independent replicates. Instead, for each patch, stratum, and solver, we summarize the pixel-level errors by a single median value, using median RAE for rainy pixels and median AE for non-rainy pixels. The median is used because within-patch error distributions are often asymmetric and can have long upper tails or rare large errors, so it provides a robust summary of typical performance within each patch.

We partitioned the pixels in every patch into rainy and non-rainy pixels according to the ground-truth using a threshold of 0.6 mm h^{-1}. For rainy pixels, we compute the relative absolute error, and for non-rainy pixels, we compute the absolute error. We summarize the medians of these results per patch after binning pixels by $d_3(p)$. Figures 4 and 5

Solver	attenuation MAE ($\mathrm{dB\,km^{-1}}$)	J_{atten}
IDW	0.050 ± 0.034	0.053 ± 0.136
ILDW	0.014 ± 0.009	$4.2 \times 10^{-3} \pm 1.4 \times 10^{-2}$
SOLVER(ILDW)	$2.1 \times 10^{-3} \pm 3.7 \times 10^{-3}$	$3.0 \times 10^{-4} \pm 1.5 \times 10^{-3}$
CONVEX SOLVER	$1.1 \times 10^{-3} \pm 9.9 \times 10^{-4}$	$3.4 \times 10^{-5} \pm 1.4 \times 10^{-4}$
HOMOTOPY SOLVER	$7.4 \times 10^{-4} \pm 6.7 \times 10^{-4}$	$1.3 \times 10^{-5} \pm 5.3 \times 10^{-5}$
SOLVER(GT)	$1.3 \times 10^{-3} \pm 1.7 \times 10^{-3}$	$5.0 \times 10^{-5} \pm 1.5 \times 10^{-4}$

Table 11
Patch-averaged attenuation error and attenuation cost-function term. Values are mean ± SD across patches. Lower values indicate better agreement with observed link attenuations.

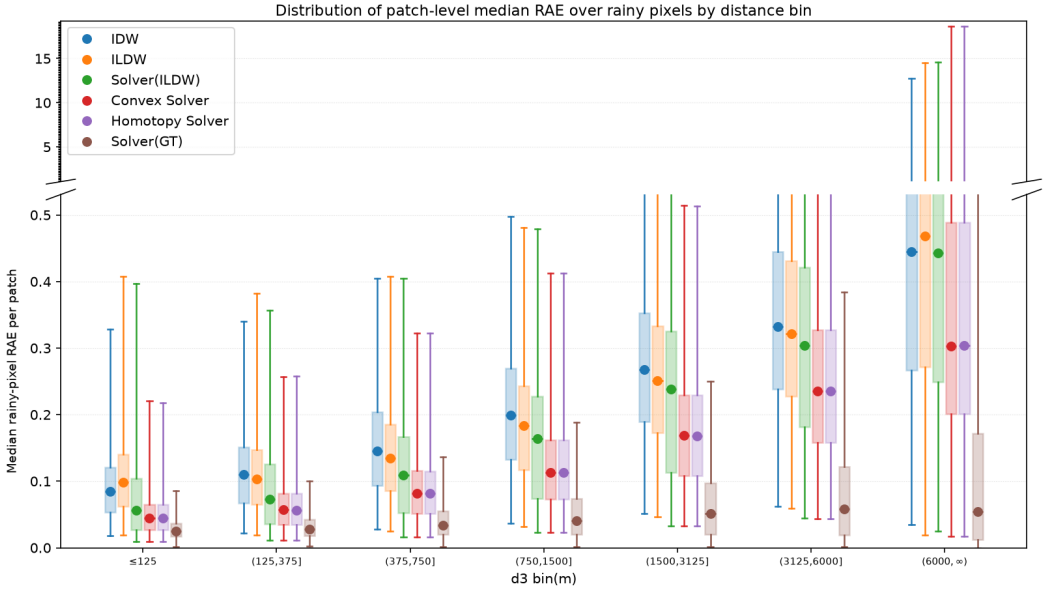


Figure 4: Rainy-pixel d_3-profile box-and-whisker summary. Each patch contributes one median rainy-pixel relative absolute error value per d_3 bin. Across patches, each box spans the 25th to 75th percentiles, and the whiskers show the minimum and maximum values. The corresponding rainy-pixel counts per bin are reported in Table 5.

depict the resulting medians for various solvers. Table 12 reports the corresponding patchwise win counts for CONVEX SOLVER relative to IDW. CONVEX SOLVER has lower median RAE/AE than IDW in a vast majority of the patches. Some patches have no pixels in strata with large d_3 distances.

8. Case Study Figures and Objective Evolution

In this section, we present two case studies, in which we compare IDW, SOLVER(GT), and CONVEX SOLVER. In the first case study, SOLVER(GT) almost equals the ground-truth while CONVEX SOLVER slightly improves over IDW.

To quantify how often these two qualitative patterns occur, we use the simple ratio

$$q(P) \triangleq \frac{\mathrm{RMSE}_{P,\mathrm{CONVEX\ SOLVER}}}{\mathrm{RMSE}_{P,\mathrm{SOLVER(GT)}}}.$$

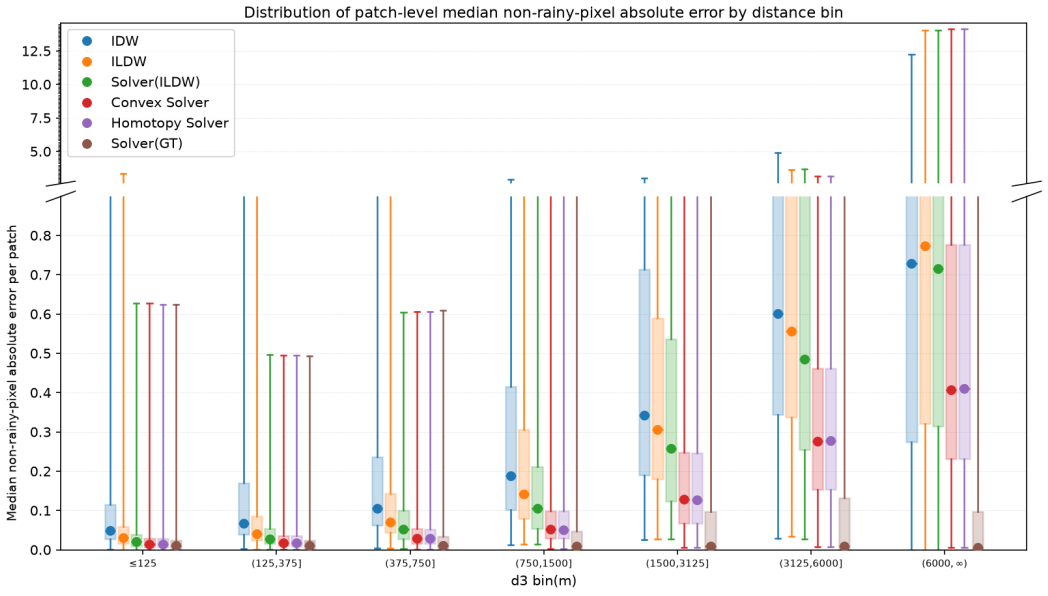


Figure 5: Non-rainy-pixel d_3-profile box-and-whisker summary. Each patch contributes one median non-rainy-pixel absolute error value per d_3 bin. Across patches, each box spans the 25th to 75th percentiles, and the whiskers show the minimum and maximum values. The corresponding non-rainy-pixel counts per bin are reported in Table 5.

d_3 bin (m)	Rainy pixels, median RAE	Non-rainy pixels, median AE
≤ 125	100 : 0	76 : 0
$(125, 375]$	100 : 0	80 : 1
$(375, 750]$	100 : 0	82 : 0
$(750, 1500]$	100 : 0	83 : 0
$(1500, 3125]$	100 : 0	86 : 0
$(3125, 6000]$	100 : 0	80 : 4
$(6000, 9000]$	83 : 17	50 : 15
$(9000, \infty)$	38 : 9	17 : 8

Table 12
Patchwise win counts by d_3-bin. Entries report CONVEX SOLVER:IDW wins. Rainy pixels are compared using median RAE, while non-rainy pixels are compared using median AE. Missing counts occur when a patch has no pixels in the corresponding bin.

We assign patch P to Case 1 (the Case Study I pattern) when $q(P) \geq 2$ and to Case 2 (the Case Study II pattern) when $q(P) < 2$. Of the 100 benchmark patches, 24 are in Case 1 and 76 are in Case 2. The smallest Case 1 ratio is 2.6310, whereas the largest Case 2 ratio is 1.9767; thus no observed patch lies immediately at the threshold.

For this patch, $J(\text{SOLVER(GT)}) = 1.485$ and $\text{RMSE}(\text{SOLVER(GT)}) = 0.102$, while $J(\text{CONVEX SOLVER}) = 0.265$ and $\text{RMSE}(\text{CONVEX SOLVER}) = 0.702$. This example shows that a lower value of the objective does not necessarily imply a more accurate reconstruction of the ground-truth rainfall field. For Case Study I, Figure 6 shows the reconstructed rainfall fields and relative errors, and Figure 7 shows the evolution of the weighted objective over L-BFGS-B iterations.

In the second case study, SOLVER(GT) and CONVEX SOLVER are visually similar in the reconstruction panels. For this patch, $J(\text{SOLVER(GT)}) = 0.289$ and $\text{RMSE}(\text{SOLVER(GT)}) = 0.552$, while $J(\text{CONVEX SOLVER}) = 0.290$ and $\text{RMSE}(\text{CONVEX SOLVER}) = 0.565$. This example shows a case in which the two solutions have almost the same

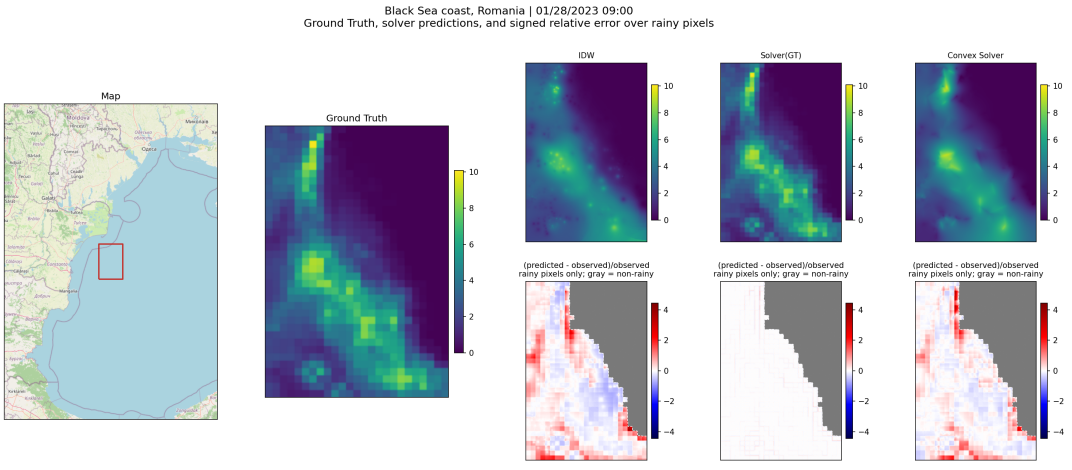


Figure 6: Case Study I: SOLVER(GT) is visually close to the ground truth. The three upper-right panels show the rainfall fields reconstructed by IDW, SOLVER(GT), and CONVEX SOLVER. The corresponding lower-right panels show the relative error of each reconstruction.

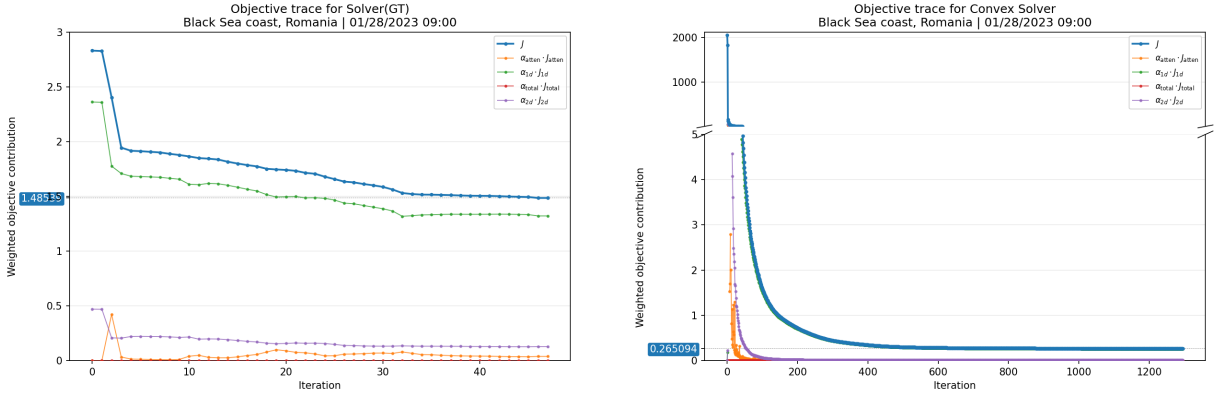


Figure 7: Evolution of the weighted objective over L-BFGS-B iterations for Case Study I. Left: SOLVER(GT), initialized from the ground-truth field, converges to final $J = 1.485$. Right: CONVEX SOLVER reaches a lower final weighted objective, $J = 0.265$, but its reconstruction is farther from the ground truth by RMSE.

objective value and almost the same reconstruction error. For Case Study II, Figure 8 shows the reconstructed rainfall fields and relative errors, and Figure 9 shows the evolution of the weighted objective over L-BFGS-B iterations.

9. Discussion

The benchmark isolates the geometric inversion problem and thereby provides a clear setting for method development. At the same time, several sources of uncertainty remain outside the scope of the current construction, including wet-antenna effects, baseline estimation, quantization, mixed-phase precipitation, and temporal dynamics. These factors matter in operational deployments and should be incorporated in later stages once the benchmarked inverse methods are well understood.

The proposed framework also offers interpretability advantages over purely data-driven alternatives. Because the objective function makes the tradeoff between attenuation consistency, spatial smoothness, and rainfall shrinkage explicit, the resulting reconstructions can be inspected and explained in terms of physically meaningful quantities.

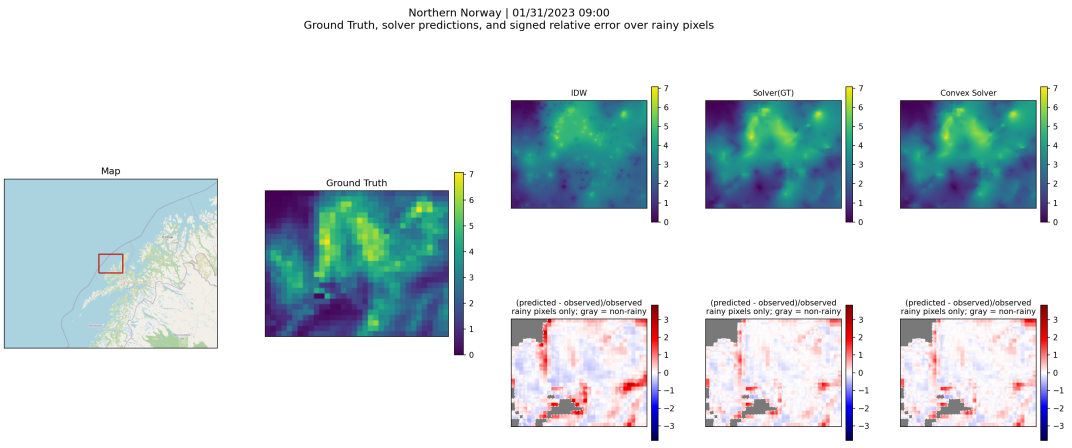


Figure 8: Case Study II: SOLVER(GT) and CONVEX SOLVER produce visually similar reconstructions. The three upper-right panels show the rainfall fields reconstructed by IDW, SOLVER(GT), and CONVEX SOLVER. The corresponding lower-right panels show the relative error of each reconstruction.

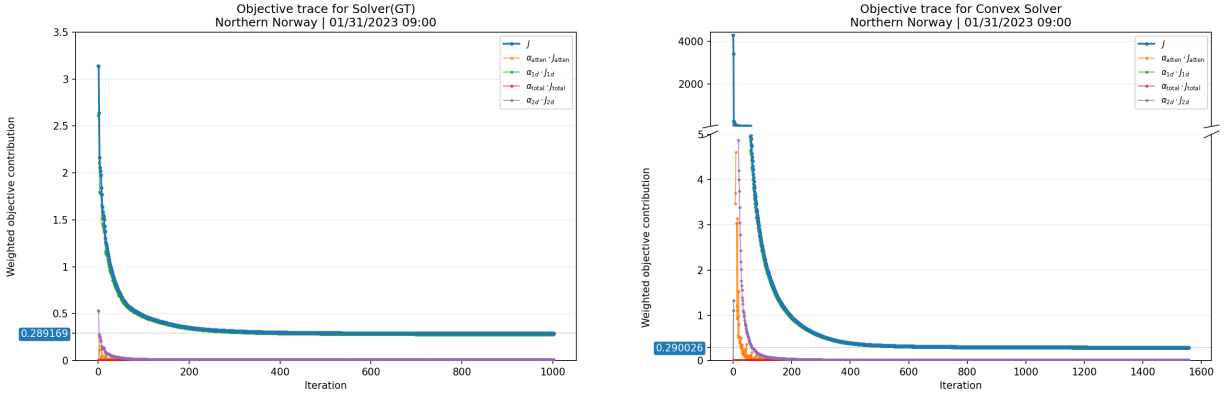


Figure 9: Evolution of the weighted objective over L-BFGS-B iterations for Case Study II. Left: SOLVER(GT), initialized from the ground-truth field, converges to final $J = 0.289$. Right: CONVEX SOLVER reaches a very similar final weighted objective, $J = 0.290$, and a similar RMSE.

The resulting optimization problem is nonconvex due to the nonlinear attenuation model. We therefore compute a locally optimal solution using L-BFGS-B, initialized from the ILDW estimate. In practice, the combination of strong data constraints and regularization yields stable and consistent reconstructions across instances.

The results highlight the importance of the initialization for the nonconvex optimization procedure. In many instances, the ILDW solution lies close to a stationary point of the objective, and the optimizer produces only marginal changes. This indicates that ILDW already achieves a strong balance between attenuation consistency and spatial regularity. Consequently, the role of the nonconvex optimization is primarily to refine the estimate in instances where these objectives are in stronger conflict. In contrast, IDW provides a weaker initialization due to its simplified geometric representation of links, leading to larger discrepancies with the attenuation model.

10. Conclusion

This journal draft integrates a benchmark-oriented view of rainfall reconstruction from commercial microwave links. Starting from hourly EURADCLIM maps, we extract rain events, simulate CML attenuations through a forward

model, and evaluate inverse reconstruction methods against known ground truth. The resulting setup reframes rainfall reconstruction from opportunistic sensing networks as a well-posed inverse problem with an explicit and interpretable objective.

References

Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyou Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. doi: 10.1137/0916069.

Feng-Wen Chen and Chen-Wuing Liu. Estimation of the spatial rainfall distribution using inverse distance weighting (idw) in the middle of taiwan. *Paddy and Water Environment*, 10(3):209–222, 2012.

Christian Chwala and Harald Kunstmann. Commercial microwave link networks for rainfall observation: assessment of the current status and future challenges. *WIREs Water*, 6(2):e1337, 2019. doi: 10.1002/wat2.1337.

Adam Eshel, Jonatan Ostrometzky, Shani Gat, Pinhas Alpert, and Hagit Messer. Spatial reconstruction of rain fields from wireless telecommunication networks–scenario-dependent analysis of IDW-based algorithms. *IEEE Geoscience and Remote Sensing Letters*, 17(5):770–774, 2020. doi: 10.1109/LGRS.2019.2935348.

Adam Eshel, Hagit Messer, Harald Kunstmann, Pinhas Alpert, and Christian Chwala. Quantitative analysis of the performance of spatial interpolation methods for rainfall estimation using commercial microwave links. *Journal of Hydrometeorology*, 22(4):831–843, 2021.

EUMETNET OPERA. Weather radar information model for implementation with the HDF5 format (ODIM_H5). https://www.eumetnet.eu/wp-content/uploads/2026/01/ODIM_H5_v2.4.2_final.pdf, 2026.

Martin Fencl, Jörg Rieckermann, Petr Sỳkora, David Stránskỳ, and Vojtěch Bareš. Commercial microwave links instead of rain gauges: fiction or reality? *Water Science and Technology*, 71(1):31–37, 2015.

Shani Gat and Hagit Messer. A comparative study of the performance of parameter estimation of a 2-d field using line-and point-projection sensors. In *2019 IEEE 8th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pages 146–150. IEEE, 2019.

Kenrich Lewis Stuart Gunn and Thomas William Russell East. The microwave properties of precipitation particles. *Quart. J. Roy. Meteorol. Soc.*, 80(346):522–545, 1954.

Jin Guo-Dong, Liu Yan-cong, and Niu Wen-Jie. Comparison between inverse distance weighting method and kriging. *Journal of Changchun University of Technology*, 3, 2003.

Congzheng Han, Gaoyuan Zhang, Baofeng Ji, Juan Huo, and Hagit Messer. On the potential of using emerging microwave links for city rainfall monitoring. *IEEE Communications Magazine*, 61(11):174–180, 2023.

ITU-R. Specific attenuation model for rain for use in prediction methods. Recommendation P.838-3, International Telecommunication Union, 2005.

Roy Janco, Jonatan Ostrometzky, and Hagit Messer. Rain estimation from smart city's e-band links. In *2022 IEEE 14th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)*, pages 1–5. IEEE, 2022.

KNMI. EURADCLIM: the european climatological high-resolution gauge-adjusted radar rainfall dataset. https://www.knmi.nl/research/observations-data-technology/projects/euradclim-the-european-climatological-high-resolution-gauge-adjusted-radar-rai 2023. Project page.

H Leijnse, R Uijlenhoet, and JNM Stricker. Microwave link rainfall estimation: Effects of link length and frequency, temporal sampling, power resolution, and wet antenna attenuation. *Advances in Water Resources*, 31(11):1481–1493, 2008.

Roberto Nebuloni, Filippo Giannetti, Fabiola Sapienza, Vincenzo Lottici, Elisa Adirosi, Giacomo Roversi, Elia Covi, Christian Gianoglio, Matteo Colli, and Carlo De Michele. A review of technical aspects and challenges in opportunistic rainfall estimation using satellite and terrestrial microwave links: How wireless infrastructure can be used for rainfall monitoring. *IEEE Geoscience and Remote Sensing Magazine*, 2025.

R. L. Olsen, D. V. Rogers, and D. B. Hodge. The aRb relation in the calculation of rain attenuation. *IEEE Transactions on Antennas and Propagation*, 26(2):318–329, 1978. doi: 10.1109/TAP.1978.1141845.

Jonatan Ostrometzky and Hagit Messer. Opportunistic weather sensing by smart city wireless communication networks. *Sensors*, 24(24):7901, 2024.

A. Overeem, H. Leijnse, and R. Uijlenhoet. Country-wide rainfall maps from cellular communication networks. *Proceedings of the National Academy of Sciences*, 110.8:2741–2745, 2013.

Aart Overeem, Hidde Leijnse, and Remko Uijlenhoet. Four-year commercial microwave link dataset for the netherlands, 2016.

Aart Overeem, Hidde Leijnse, and Remko Uijlenhoet. EURADCLIM: the european climatological high-resolution gauge-adjusted radar rainfall dataset and its potential for understanding extreme precipitation. *Earth System Science Data*, 15:1441–1460, 2023. doi: 10.5194/essd-15-1441-2023.

Pauli Virtanen et al. Scipy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, 17:261–272, 2020.

Ciyou Zhu, Richard H. Byrd, Jorge Nocedal, and José Luis Morales. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software*, 23(4):550–560, 1997. doi: 10.1145/279232.279236.

Dale Zimmerman, Claire Pavlik, Amy Ruggles, and Marc P Armstrong. An experimental comparison of ordinary and universal kriging and inverse distance weighting. *Mathematical Geology*, 31(4):375–390, 1999.

A. Illustrative Examples

This appendix presents a minimal example that isolates an important property of the inverse problem. Namely, we give a concrete counterexample showing that J_{atten} is nonconvex.

A.1. A 3×3 Counterexample to Convexity

Consider one horizontal link ℓ crossing the three middle-row pixels of a 3×3 grid. The crossed length is 0.125 km in each pixel, so $|\ell| = 0.375$ km. Let the link use horizontal polarization at 38.003 GHz, for which the ITU-R coefficients used here are $\kappa = 0.400171594750$ and $\alpha = 0.881535333268$. For this one-link instance,

$$\widehat{A}(R) = \sum_{p \in \ell} d_p \kappa R(p)^{\alpha}, \qquad J_{\text{atten}}(R) = \frac{\left(\widehat{A}(R) - A_{\text{obs}}\right)^2}{|\ell|}.$$

Take the two strictly positive fields

$$R^{(L)} = \begin{bmatrix} 1 & 1 & 1 \\ 20 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad R^{(R)} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 20 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{mm h}^{-1}.$$

They are left-right reflections across the grid's vertical centerline. Either field could be the ground truth: the link crosses the reflected pixels over equal lengths, so both fields produce

$$\widehat{A}\left(R^{(L)}\right) = \widehat{A}\left(R^{(R)}\right) = A_{\text{obs}} = 0.801595407171 \text{ dB},$$

and hence $J_{\text{atten}}(R^{(L)}) = J_{\text{atten}}(R^{(R)}) = 0$. Their arithmetic mean is

$$R^{(M)} = \frac{R^{(L)} + R^{(R)}}{2} = \begin{bmatrix} 1 & 1 & 1 \\ 10.5 & 1 & 10.5 \\ 1 & 1 & 1 \end{bmatrix}.$$

For $g(r) = r^{\alpha}$, we have $g''(r) = \alpha(\alpha-1)r^{\alpha-2} < 0$ because $0 < \alpha < 1$; hence g is strictly concave and $2(10.5)^{\alpha} > 20^{\alpha} + 1$. Consequently,

$$\widehat{A}(R^{(M)}) = 0.845082934781 \text{ dB} > A_{\text{obs}}, \qquad J_{\text{atten}}(R^{(M)}) = 0.005043106820 > 0.$$

Therefore,

$$J_{\text{atten}}\left(\frac{R^{(L)} + R^{(R)}}{2}\right) > \frac{J_{\text{atten}}(R^{(L)}) + J_{\text{atten}}(R^{(R)})}{2} = 0,$$

which violates the defining inequality for a convex function.